

Lattice — Feature Showcase

Open this file with Lattice to see every feature rendered live.
Switch between **Edit**, **Preview**, and **Dual** view with the toolbar selector.
Try `Ctrl/Cmd+Shift+T` to auto-generate the Table of Contents below.

Table of Contents

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Text Formatting

Normal paragraph text flows here. Lattice renders Markdown as **close to GitHub** as possible, so files look identical whether viewed in the editor preview, on GitHub, or printed.

Style	Syntax	Result
Bold	**bold**	bold
Italic	<i>*italic*</i>	<i>italic</i>
Bold + italic	***both***	<i>both</i>
Strikethrough	~~strikethrough~~	strikethrough
Inline code	<code>`code`</code>	code
Superscript HTML	X²	X²
Subscript HTML	H₂O	H₂O

Highlight Marks

Highlight marks are a Lattice-exclusive feature. Wrap any text in `==double equals==` to render it with a **bright amber background**.

This works in **any context** — inside a sentence, `==across italic and bold spans==`, or on its own line:

This entire sentence is highlighted.

[!NOTE] Highlight marks are **not** rendered inside code blocks or inline code — `==this stays raw==` .

Lists

Unordered

- First item
- Second item
 - Nested item A
 - Nested item B
 - Deeply nested
- Third item with **bold** and *italic* text

Ordered

1. Install Lattice from the [releases page](#)
2. Open any `.md` file — double-click or via **File** → **Open**
3. Choose your view:
 - i. **Edit** — raw CodeMirror editor only
 - ii. **Preview** — rendered HTML only
 - iii. **Dual** — side-by-side (four layout variants)
4. Print with `Ctrl/Cmd+P` — always renders to a clean light theme

Definition-style (using bold + indent)

Local-first All files stay on your machine. No cloud sync required.

Documents as Code Plain-text `.md` files are diff-friendly, version-controlled, and AI-ready.

Task Lists (GFM)

GitHub Flavored Markdown task lists render as interactive checkboxes in Lattice.

- ☒ Install Lattice
- ☒ Open the demo file
- ☒ Explore Edit / Preview / Dual views
- ☐ Try `Ctrl/Cmd+Shift+T` to regenerate the Table of Contents
- ☐ Try `Ctrl/Cmd+Shift+L` to auto-align a pipe table
- ☐ Paste a screenshot with `Ctrl/Cmd+V` — it auto-saves to a sibling folder
- ☒ Open a second file in a new window (multi-window support)
- ☐ Try printing to PDF with `Ctrl/Cmd+P`

Tables

Lattice supports **GFM pipe tables** with left, center, and right alignment. Press `Ctrl/Cmd+Shift+L` with the cursor anywhere in the table to auto-format it.

Feature	Edit View	Preview View	Dual View
CodeMirror editor	✓	—	✓
Live rendered preview	—	✓	✓
Scroll sync	—	—	✓
Print (Ctrl+P)	✓	✓	✓
Highlight marks	✓	✓	✓
Mermaid diagrams	—	✓	✓
KaTeX mathematics	—	✓	✓

Code

Inline code

Use backticks for `inline code`. Great for `file/paths`, `commands`, and `variable_names`.

Fenced code blocks

Lattice passes code fences to the preview unchanged — syntax highlighting is provided by the renderer.

```
// Lattice highlight-mark plugin – converts ==text== to <mark> elements
export const MARK_RE = /==(?:[^\n]|(?:!|=))+?)/g;

export function splitAtMarks(text: string): HastNode[] {
  const parts: HastNode[] = [];
  let lastIndex = 0;
  MARK_RE.lastIndex = 0;
  let match: RegExpExecArray | null;
  while ((match = MARK_RE.exec(text)) !== null) {
    if (match.index > lastIndex)
      parts.push({ type: 'text', value: text.slice(lastIndex, match.index) });
  }
}
```

```

    parts.push({
      type: 'element', tagName: 'mark',
      properties: {},
      children: [{ type: 'text', value: match[1] }],
    });
    lastIndex = match.index + match[0].length;
  }
  if (lastIndex < text.length)
    parts.push({ type: 'text', value: text.slice(lastIndex) });
  return parts;
}

// Tauri command — read file with conflict detection
#[tauri::command]
pub async fn read_file(path: String) -> Result<FileReadResult, String> {
  let content = fs::read_to_string(&path)
    .map_err(|e| format!("Cannot read {path}: {e}"));
  let hash = compute_sha256(&content);
  Ok(FileReadResult { content, hash })
}

# Build a production release
export LATTICEBUILD_NO="MyBuild-001"
npm run tauri build

```

Blockquotes

Single-level blockquote. Great for callouts, citations, or side notes.

Nested blockquotes are also supported:

Inner quote — *"Documents as code, not documents as data."*

— Lattice design philosophy

GitHub-style Alerts (rendered by GitHub and Lattice)

[!NOTE] Useful information that users should know, even when skimming.

[!TIP] Optional advice for a better experience.

[!IMPORTANT] Key information users need to know to achieve their goal.

[!WARNING] Urgent information that needs immediate attention.

[!CAUTION] Advises about risks or negative outcomes.

Mathematics — KaTeX

Lattice renders mathematics via **KaTeX**. Use $\dots$ for inline and $\displaystyle \dots$ for display math.

Inline math

Einstein's famous equation $E = mc^2$ and the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.

The Euler identity $e^{i\pi} + 1 = 0$ is often called the most beautiful equation in mathematics.

Display math — Calculus

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$$

$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Display math — Linear Algebra

$$\mathbf{Ax} = \lambda \mathbf{x} \implies \det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Display math — Series & Limits

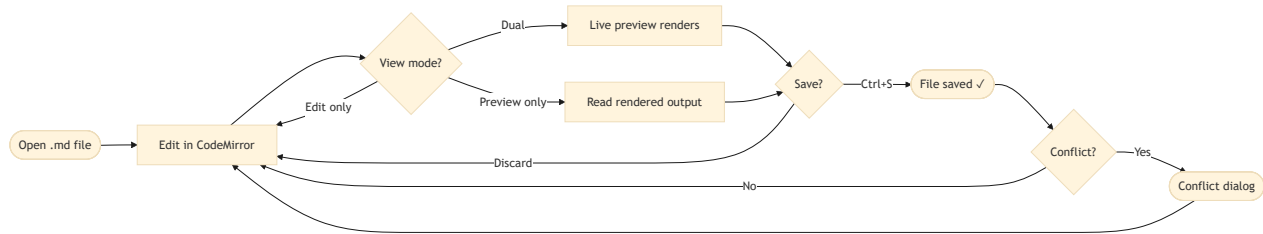
$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n = e$$

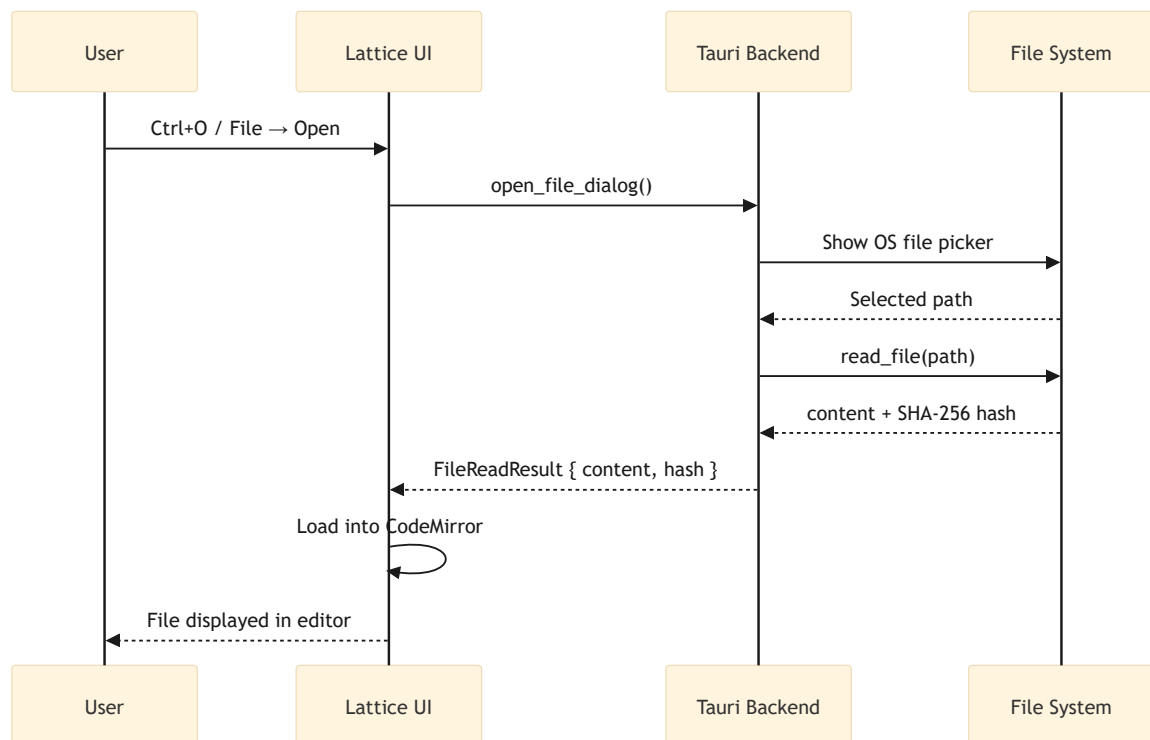
Mermaid Diagrams

Lattice renders ````mermaid```` fenced blocks inline in the preview.

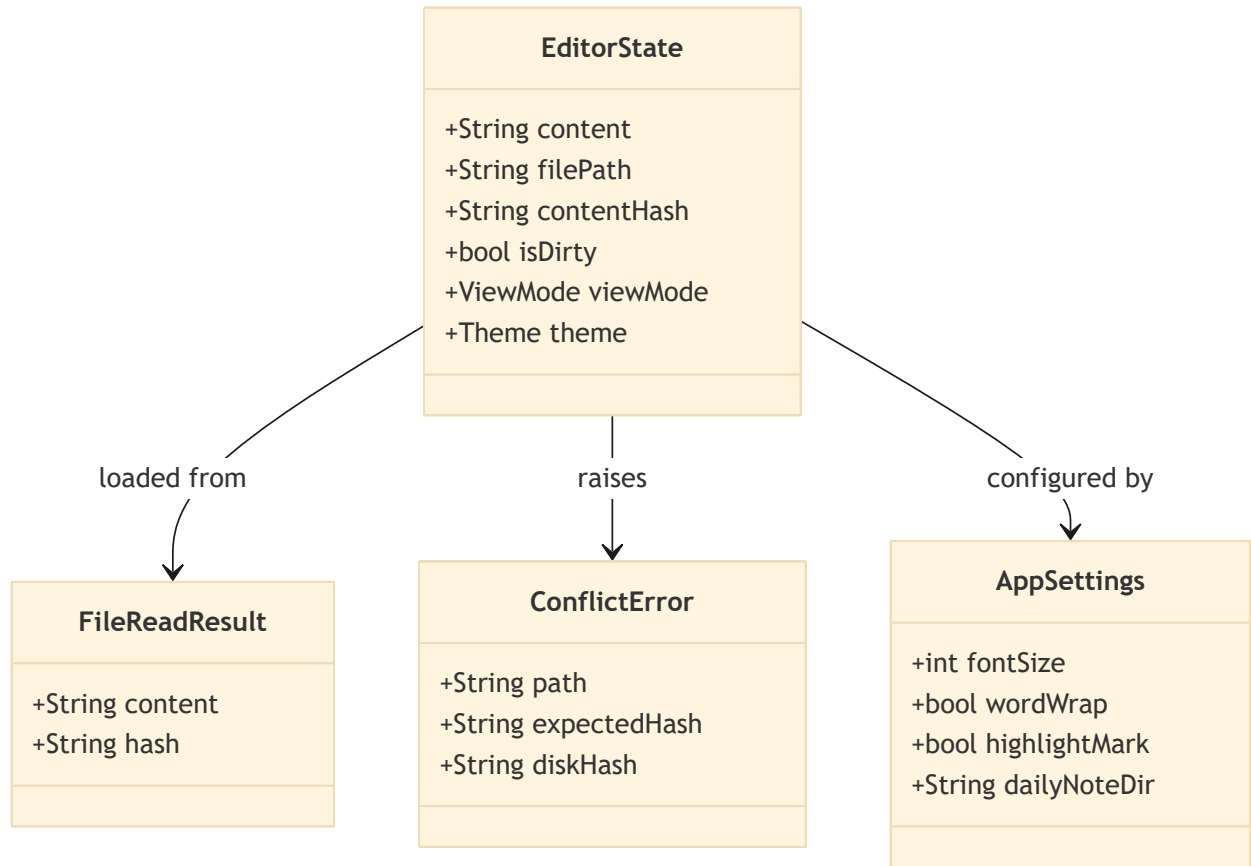
Flowchart — Lattice Edit Cycle



Sequence Diagram — File Open Flow

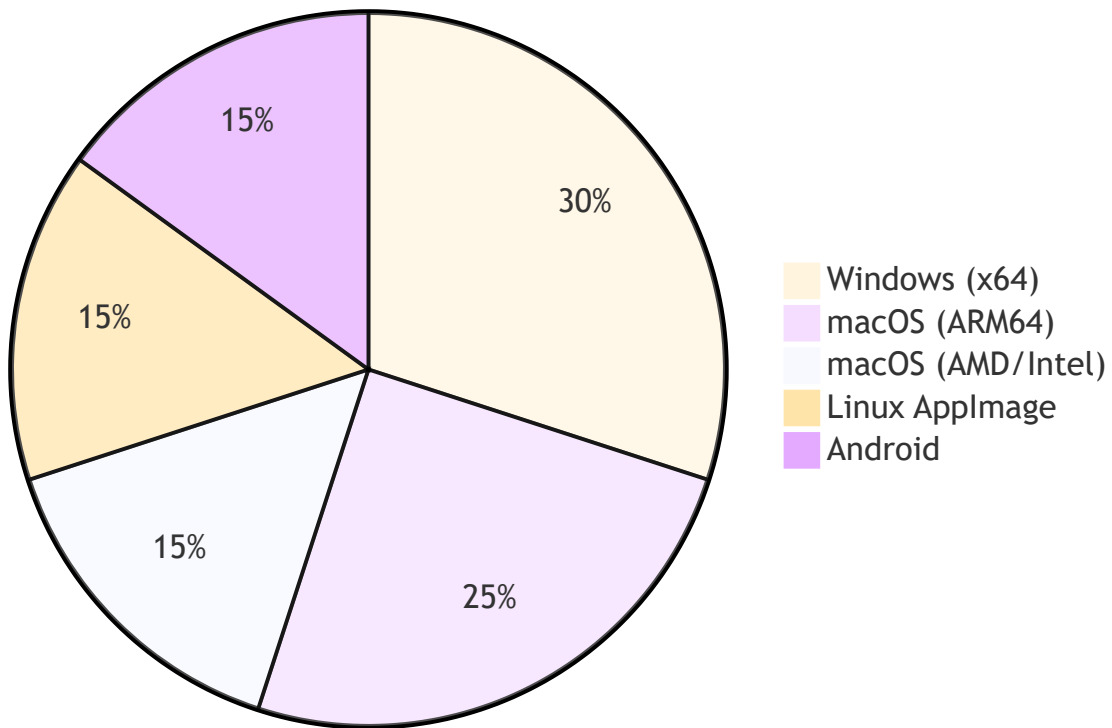


Class Diagram — Core Data Model



Pie Chart — Supported Platforms

Lattice Platform Coverage



Images

Paste images directly into Lattice with `ctrl/cmd+v`.

The image file is automatically saved to a sibling folder named after the current document, and a Markdown reference is inserted at the cursor position:

```
![Image](demo_assets/img_1780574419628.png)
```

```
![Image](demo_assets/img_1780575215039.png)
```

Here is how it renders in `Dual (edit on the left) mode`:

demo.md - lattice (v0.2.39 - windows-desktop-2026W23.0EEC)
90% Dual (edit on the left)

```

207 ## Mathematics — KaTeX
208
209 Lattice renders mathematics via KaTeX. Use '$...$' for inline and
    '$$...$$' for display math.
210
211 ### Inline math
212
213 Einstein's famous equation $E = mc^2$ and the quadratic formula
214 $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ render inline.
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218 ### Display math — Calculus
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221 \int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}
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245 e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots
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248
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250 \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e
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252
253 ---
254

```

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Mermaid Diagrams

Lattice renders `mermaid` fenced blocks inline in the preview.

Flowchart — Lattice Edit Cycle

```

graph TD
    Open[Open and file] --> Edit[Edit in CodeMirror]
    Edit -- Save --> Live[Live preview renders]
    Edit -- Preview only --> Read[Read rendered output]
    Edit -- Discard --> Edit
    Live --> Send[Send]
    Send --> Parse[Parse and J]
    Parse --> Conflict{Conflict?}
    Conflict -- Yes --> Dialog[Conflict dialog]
    Conflict -- No --> Edit
    Read --> Edit

```

Here is how it renders in **Dual** (edit on the bottom) mode:

233 $\det(\mathbf{A}) - \lambda \det(\mathbf{L}) = 0$

Tip: The assets folder name is derived from the document filename, keeping related files grouped together and easy to version-control.

Links

- [Lattice on GitHub](#) — source code and issue tracker
- [Releases Page](#) — download installers
- [Release Site](#) — GitHub Pages release portal
- Relative link: [System Architecture Demo](#)
- Relative link: [Physics Equations Demo](#)

Auto-link: <https://github.com/blessia-blessini/lattice>

HTML in Markdown

Lattice passes safe inline HTML through to the preview. Useful for fine-grained formatting unavailable in pure Markdown:

`<details> <summary>Click to expand — collapsible section</summary>`

Hidden content revealed on click. Great for long appendices or spoilers.

```
{
  "product": "Lattice",
  "version": "0.3.0",
  "license": "AGPL-3.0"
}
```

`</details> <br>`

HTML element	Use case
<code><mark></code>	Rendered by <code>==highlight==</code> syntax
<code><details></code>	Collapsible sections
<code><sup></code>	Superscripts: X <code><sup>2</sup></code>
<code><sub></code>	Subscripts: CO <code><sub>2</sub></code>

HTML element	Use case
 	Explicit line break
<kbd>	Keyboard keys: <kbd>Ctrl</kbd> + <kbd>S</kbd>

Keyboard Shortcuts Reference

Action	Windows / Linux	macOS
Save	<kbd>Ctrl+S</kbd>	<kbd>Cmd+S</kbd>
Open file	<kbd>Ctrl+O</kbd>	<kbd>Cmd+O</kbd>
New window	<kbd>Ctrl+N</kbd>	<kbd>Cmd+N</kbd>
Generate / refresh TOC	<kbd>Ctrl+Shift+T</kbd>	<kbd>Cmd+Shift+T</kbd>
Auto-align table	<kbd>Ctrl+Shift+L</kbd>	<kbd>Cmd+Shift+L</kbd>
Paste image from clipboard	<kbd>Ctrl+V</kbd>	<kbd>Cmd+V</kbd>
Print / Save as PDF	<kbd>Ctrl+P</kbd>	<kbd>Cmd+P</kbd>
Increase font size	<kbd>Ctrl++</kbd>	<kbd>Cmd++</kbd>
Decrease font size	<kbd>Ctrl+-</kbd>	<kbd>Cmd+-</kbd>